Lamp lab: adult guide to the compact catalog

For week-02-shared-catalog.pdf (F02-S-CAT-v2, Problems 1–10) with the boards and counters in week-02-shared-catalog-boards.pdf. Unpiloted. Every answer below is replayed by plans/verify-week-02-catalog-kit.py.

Mathematical overview

The move. Pressing a line flips both lamps at its ends, OFF to ON or ON to OFF. Pressing the same line again undoes it, and the order of presses does not matter. So any run of presses does what the set of lines pressed an odd number of times does, and a puzzle asks for a set of lines that touches each lamp to be changed an odd number of times and every other lamp an even number of times. The lamps to be changed are the ones that differ between start and target.

Theorem. A target can be made from a start exactly when every island (connected piece of the board) has an even number of lamps that differ between start and target.

Proof. An odd island is impossible because a press changes two lamps of one island, so the number of ON lamps on each island changes by 0 or 2 and stays odd or even for ever (the invariant). Every even case works by path pairing: on one island, pair up the lamps to be changed and press every line along a path joining each pair. The two ends flip once; each lamp inside the path flips twice, which is no change. For adults: states are vectors over the field with two elements, a press adds the two end lamps, and the reachable changes are exactly the vectors with an even sum on every island.

A walking light is the special case children meet first: with one lamp ON, pressing the lines of a path from it moves the light to the path's far end (Problems 2–4). From all OFF the same presses light both ends (Problem 5).

Assumptions and limits. Both lamps flip on every press. A child who changes one lamp breaks the invariant without noticing, which is what happened when the rule lived only on paper; hence counters and a partner who flips both. Solutions are not unique on a board with a loop (square, ring, grid), since pressing every line of a loop changes nothing; on the tree a reachable target has exactly one set of lines. Shortest solutions belong to the upper catalog.

What children are meant to find. Experiments: Problems 1–7, where every target can be made and a press is undone by pressing again. Conjecture: Problem 8, after some tries, “it can't be done.” Explanation: the left island starts with one light and every press changes two of its lamps, so that light can never go out. Problem 9 tests the explanation, since any bridge makes the trip possible, and Problem 10 makes children the puzzle makers.

Band map. K–1: Problems 1–4 and 6 with counters, the adult reading each prompt; Problem 8 as a short “can you?” Grades 2–3: all ten, with Problems 8–10 the core. Grades 4–5: the upper catalog (week-02-shared-catalog-upper.pdf and plans/week-02-catalog-upper.md), with Problems 8–9 here as an optional five-minute start.

Materials and preparation

Catalog: one per child (4 pages). Its small pictures are the start and target cards. **Boards:** sheets 1–3 of week-02-shared-catalog-boards.pdf on copy paper, one set per pair: K–1 2, grades 2–3 2, grades 4–5 1 and one for the launch, 18 sheets. **Counters:** sheet 4 on card stock, one sheet cut into six sets of 11 with a paper trimmer. Dot up is ON. Two-colour counters already on hand also work; choose one colour for ON. **Also:** pencils and erasers for the Problem 9 bridge and scrap paper for Problem 10. **Preparation:** about 20 minutes. The boards and counters are digitally checked but physically unrehearsed: run the launch once on a board before class.

Launch together (about 5 minutes)

Gather everyone at the adults' square board, all counters blank up. (1) “Each counter is a lamp. Dot up is ON.” Let children turn a few over, then set all OFF again. (2) **Press:** one child points to a line and a partner flips both counters at its ends; two lamps are ON. (3) **Press again:** point to the same line and flip both again; all OFF. Pressing twice undoes. (4) **One target:** show Problem 1's first pair, start all OFF and target top and right ON, and ask which press makes it. Then hand out catalogs and boards. Pointer and flipper swap after each press; a third child checks that both counters flipped. Laying out a start picture is setup, not a press.

The one-light rule

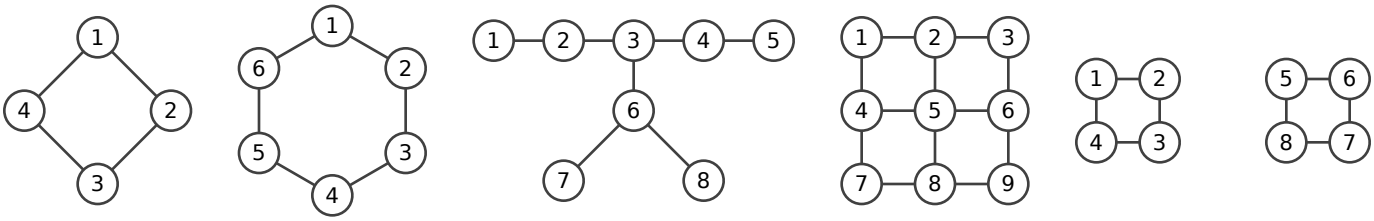
The catalog dropped “Keep exactly ONE lamp ON” from Problems 2–4. Problems 2 and 4 stand as printed: any route counts, and the answers below keep one light anyway. Problem 3 needs the rule: “the light” means one light, and without the rule the tour can be done by lighting lamps and undoing them (from lamp 1: 2-3, 4-5, 6-1, 6-1, 4-5, 2-3 lights every lamp at some point and ends with lamp 1 alone). When you read Problem 3 aloud, add: “Keep exactly one lamp ON after every press.”

Testing an invented puzzle (Problem 10)

On each island, count the lamps that differ between the start and the target. If every count is even, the puzzle can be made; if any count is odd, it cannot. The old shared guide promised that “reversing the moves always succeeds.” That holds only for a puzzle made by pressing lines from the start. A drawn target can be impossible: on the square, all OFF to one lamp ON; on the islands, all OFF to one lamp ON on each island, although two lamps change in all. Ask the maker to make the target by presses first, or to apply the count.

Answers

Lamp numbers follow these keys; the children's boards are unnumbered. “1-2, 2-3” means press line 1-2, then line 2-3, and {1, 3} names the target with lamps 1 and 3 ON. Reset to the printed start for each pair of pictures. Any working list counts; these are examples.



square **six-ring** **tree** **3 × 3 grid** **islands**

Problem 1 (square, all OFF to start). {1, 2}: 1-2. {1, 3}: 1-2, 2-3. {1, 2, 3, 4}: 1-2, 3-4.

Problem 2 (six-ring, lamp 1 ON to start). {3}: 1-2, 2-3. {4}: 1-2, 2-3, 3-4. Going the other way round also works.

Problem 3 (six-ring, lamp 1 ON to start and to finish). {1}: 1-2, 2-3, 3-4, 4-5, 5-6, 6-1. One light walks round the ring and comes home; detours are fine. Read the one-light rule aloud (see above).

Problem 4 (tree, lamp 1 ON to start). {5}: 1-2, 2-3, 3-4, 4-5. {7}: 1-2, 2-3, 3-6, 6-7. {8}: 1-2, 2-3, 3-6, 6-8. Backtracking is legal.

Problem 5 (tree, all OFF to start). {1, 8}: 1-2, 2-3, 3-6, 6-8. These are the presses of the trip to 8 in Problem 4: from all OFF the starting lamp lights too, and the middle lamps flip twice. On a tree this is the only set of lines that works.

Problem 6 (six-ring, all OFF to start). {1, 2}: 1-2. {1, 4}: 1-2, 2-3, 3-4. {1, 2, 4, 5}: 1-2, 4-5. {1, 2, 3, 4, 5, 6}: 1-2, 3-4, 5-6. The ring has a loop, so other sets work too: pressing every line of the ring changes nothing.

Problem 7 (3 × 3 grid, all OFF to start). {1, 5}: 1-2, 2-5. {2, 8}: 2-5, 5-8. {1, 3, 7, 9}: 1-2, 2-3, 7-8, 8-9. {1, 3, 4, 6, 7, 9}: 1-2, 2-3, 4-7, 6-9. Pair the target lamps and join each pair by a path.

Problem 8 (islands, lamp 1 ON to lamp 5 ON). Impossible. Every press changes two lamps on one island, so the left island's number of ON lamps stays odd: its one light can never go out. Accept that reason in any words, or a convincing run of failed tries for younger children, and keep the search short.

Problem 9 (the same start and target, with one bridge). Every one of the 16 possible bridges works. With bridge 2-5: 1-2, 2-5. Each child's bridge needs its own route: walk the light to the bridge, across, and on to 5. Draw the bridge on the islands board. In the catalog the start and target pictures sit closer together than the two islands, so a child drawing there may join start to target instead.

Problem 10 (invented puzzles). See the test above. A puzzle made by pressing lines from the start can always be undone by pressing the same lines again; a target drawn freehand may be impossible.

Timing and hints

A first visit: launch 5 minutes; Problems 1–3, 15; Problems 4–7, 15; Problems 8–9, 10; Problem 10 as time allows. Stop a route after a satisfying attempt; finishing the catalog is not the goal. Hints, only when needed: “Which lamps have to change?”; “Find a path from one lamp you want ON to the other.”; for Problem 8, “Count the lights on the left island after each press.” For a pair that finishes early: “Make a target your partner cannot make, and explain why.”

Sources and status

Answers and trio roles come from the archived shared guide, F02-S-FAC-v1 (archive-before-deslopping/, old Problems 2 and 5–13), renumbered. The reversal caveat and the one-light decision answer the math check, plans/review/week-02-math.md, items 4 and 5. Lowell year-1 Handout 9, Problem 9.1, toggles doors, a related earlier activity. Adult reading: sets of lines with prescribed odd lamps are T-joins (Cornuéjols, Combinatorial Optimization: Packing and Covering, ch. 2). Status: unpiloted. A paper lamp session went poorly when the rule lived only on paper (worksheet-workflow/context.md, October 3, 2026); the counters and partner presses respond to that and are untested.